

Math 213 Recitation - Week 01

1. Let D be the region in the xy -plane given by the inequalities $0 \leq x \leq 4$, $0 \leq y \leq 4$. Consider the partition $\mathcal{P}$ obtained by dividing the square D into four equal subsquares. Approximate the double integral

$$\iint_D (x^2 + y) dA$$

- (a) by computing the Riemann sum $R(f, \mathcal{P})$ using the center point of each subsquare,
- (b) by computing the lower Riemann sum corresponding to the partition $\mathcal{P}$,
- (c) by computing the upper Riemann sum corresponding to the partition $\mathcal{P}$.

Also compare the results you have found.

2. Evaluate the following double integrals by interpreting them as volumes.

- (a) $\iint_R \mu(u, v) dA$ where R is given by $-4 \leq x \leq 1$, $-1 \leq x \leq 3$.
- (b) $\iint_D (x + 3) dA$ where D is the upper semi-disc of radius 2.
- (c) $\iint_{x^2 + y^2 \leq a} \sqrt{a^2 - x^2 - y^2} dA$
- (d) $\iint_T (1 - x - y) dA$ where T is the triangle with the vertices $(0, 0)$, $(1, 0)$, $(0, 1)$.

3. Express the volume of the solid bounded by $z = 1$ and $z = 2 - x^2 - y^2$.
4. Express the volume of the solid region below $z = x^2 + y^2$, above xy -plane and inside the cylinder $x^2 + y^2 = 2x$.
5. Evaluate

$$\iint_D (10x^2y^3 - 6) dA$$

where D is the plane region bounded by the curves $x = -2y^2$ and $x = y^3$.

6. Compute the volume of the solid region bounded by $z = 8 - x^2 - 2y^2$ and the planes $x = 0$, $y = 0$, $z = 0$ and $x + 2y = 2$.
7. Let f be a continuous function, which is therefore integrable over a bounded domain. Express the double integral $\iint_D f(x, y) dA$ as iterated integrals in both orders $dx dy$ and $dy dx$ where
 - (a) D is the plane region inside $x^2 + y^2 = 4x$ and to the left of the line $x = 3$,
 - (b) D is the plane region in the 1st-quadrant, below $x + y = 5$ and above $x + y = 4$.
8. Reverse the order of the iterated integrals below:

- (a)

$$\int_0^1 \int_{x^2}^{2-x} f(x, y) dy dx.$$

- (b)

$$\int_0^1 \int_{\sqrt{y}}^y f(x, y) dx dy.$$

Math 213 Recitation - Week 02

1. Evaluate

$$\int_0^2 \int_{y/2}^1 ye^{-x^3} dx dy.$$

2. Given a continuous function $g : \mathbb{R} \rightarrow \mathbb{R}$, define

$$h(x) = \int_0^x \int_0^y g(t) dt dy.$$

Show that

$$h(x) = \int_0^x (x-t)g(t) dt.$$

3. Let

$$f(x, y) = \begin{cases} y^{-2}, & 0 < x < y < 1 \\ -x^{-2}, & 0 < y < x < 1 \\ 0, & \text{otherwise.} \end{cases}$$

Let S be the unit square $[0, 1] \times [0, 1]$.

- (a) Show that f is not integrable on S but $f(x, y)$ is integrable on $[0, 1]$ as a function of one variable if we fix one of the variables.
(b) Show that the iterated integrals

$$\int_0^1 \int_0^1 f(x, y) dx dy \text{ and } \int_0^1 \int_0^1 f(x, y) dy dx$$

both exist and are unequal.

4. Evaluate

$$\iint_D \frac{1}{(x+y)^2}$$

where D is given by $0 \leq x \leq 1$, $0 \leq y \leq x^2$.

5. Determine whether

$$\iint_D \frac{1}{(1+x^2)(1+y^2)} dA$$

where D is the 1st-quadrant is convergent or not.

6. Determine whether

$$\iint_D \frac{1}{1+x+y} dA$$

where D is given by the inequalities $0 \leq x \leq \infty$, $0 < y < 1$ is convergent or not.

7. Determine whether

$$\iint_{[0,1]^2} \frac{1}{\sqrt{x+y}} dA$$

is convergent or not.

8. Find the average value of $f(x, y) = x^2$ over the rectangle $a \leq x \leq b$, $c \leq y \leq d$.

Math 213 Recitation - Week 03

1. Find the area of the region inside the cardioid $r = 1 + \cos\theta$.

2. Iterate the following double integrals in polar coordinates:

(a) $\iint_D \frac{1}{(x^2+y^2)^2} dA$ where $D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \geq 1, 0 \leq x, y \leq 1\}$.

(b) $\iint_D \cos(x^2 + y^2) dA$ where $D = \{(x, y) \in \mathbb{R}^2 \mid x \geq 0, 1 \leq x^2 + y^2 \leq 4\}$.

3. Consider the space region

$$E = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 \leq 16, (x+1)^2 + y^2 \leq 9\}.$$

Express the volume of E as iterated integrals in polar coordinates.

4. Evaluate $\int_0^\infty e^{-x^2} dx$.

5. Find the volume of the region above xy -plane, below the cone $z = 2 - \sqrt{x^2 + y^2}$, and inside the cylinder $(x-1)^2 + y^2 = 1$.

6. Evaluate

$$\iint_D \frac{6y^2 \sin(x^2 y^2)}{x} dA$$

where D is the region bounded by the curves $xy = 1$, $xy = 2$, $y = x^2$, and $y = 3x^2$.

7. Evaluate

$$\iint_D \frac{ye^{2y^2-x^2}}{x} dA$$

where D is the region lying in the 1st-quadrant and bounded by the curves $-x^2 + 2y^2 = 1$, $-x^2 + 2y^2 = 3$, $y = x$, $y = 4x$.

Note: Do not forget to take the sign of the Jacobian determinant into account.

8. Use the transformation $x = u - uv$, $y = uv$ to evaluate

$$\iint_S (x+y)^{-1} dA$$

where S is the region in the 1st-quadrant between the lines $x+y=1$ and $x+y=4$.

Hint: Instead of finding the boundaries for v algebraically, which seems very difficult, notice that

$$\forall u \in [1, 4] \quad \forall y \in \{(x, y) \mid x+y=u\}, \quad v = y/u \in [0, 1].$$

One can observe this by drawing a picture. And one can argue that v takes every value in $[0, 1]$ by Intermediate Value Theorem arguments. So, the boundaries for v are basically 0 and 1.

Note: It turns out that this problem is a little bit difficult. I will solve this next week on board with detailed explanations.

9. Let $f(x, y)$ be continuous function. Given $r \in \mathbb{R}$, define D_r to be disk centered at the point (a, b) with radius r . Show that

$$\lim_{r \rightarrow 0} \frac{1}{\pi r^2} \iint_{D_r} f(x, y) dA = f(a, b).$$

10. Transform to Cartesian coordinates and sketch the graphs:

- (a) $r = 3 \sec \theta$,
- (b) $r = \sin \theta + \cos \theta$,
- (c) $r = \sec \theta \tan \theta$,
- (d) $r = 2/\sqrt{\cos^2 \theta + 4 \sin^2 \theta}$,
- (e) $r = 2/(1 - \sin \theta)$.

11. Sketch the polar graph of the equation $r = 1 + \sin \theta$.

12. Find all intersection points of the following pairs of curves:

- (a) $r = \sqrt{3} \cos \theta$ and $r = \sin \theta$,
- (b) $r = 1 + \cos \theta$ and $r = 3 \cos \theta$.

Solution of P9: By MVT for double integrals; given any $r \in \mathbb{R}^+$, there exists a point $(a_r, b_r) \in D_r$ such that

$$\frac{1}{\pi r^2} \iint_{D_r} f(x, y) dA = f(a_r, b_r).$$

Moreover; whatever the points (a_r, b_r) are, we have that

$$\lim_{r \rightarrow 0^+} (a_r, b_r) = (a, b).$$

Then, by the continuity of f we obtain

$$\lim_{r \rightarrow 0} \frac{1}{\pi r^2} \iint_{D_r} f(x, y) dA = \lim_{r \rightarrow 0^+} f(a_r, b_r) = f(\lim_{r \rightarrow 0^+} (a_r, b_r)) = f(a, b).$$

An Off-topic Theoretical Note: While arguing

$$\lim_{r \rightarrow 0^+} (a_r, b_r) = (a, b),$$

we are in fact considering a function $g : \mathbb{R}^+ \rightarrow \mathbb{R}^2$ given by $g(r) = (a_r, b_r)$ with limit (a, b) at 0. Of course; given r , the point (a_r, b_r) is not necessarily unique and therefore we are not using an explicitly given function. Rather, we invoke the axiom of choice to guarantee that such a function g exists. Using the axiom of choice might not be necessary here, but we use it anyway since without the axiom of choice we need to construct a function g explicitly and such construction would probably be very tedious.

Math 213 Recitation - Week 04

1. Let T be the tetrahedron bounded by the coordinate planes and the plane $3x + 2y + z = 6$. Evaluate

$$\iiint_T (x + y + z) dV.$$

2. Let T be the triangular region in the xy -plane with vertices $(0, 0)$, $(2, 0)$, $(2, 2)$. Let Ω be the solid region

$$\Omega = \{(x, y, z) \in \mathbb{R}^3 \mid (x, y) \in T, 0 \leq z \leq 4 - x\}.$$

Write down the integral

$$I = \iiint_{\Omega} x dV$$

as iterated integrals in the order

(a) $dzdydx$,

(b) $dydzdx$.

3. Let S be the solid region in the 1st-octant, inside the sphere $x^2 + y^2 + z^2 = 9$, outside the cylinder $x^2 + y^2 = 4$. Find the volume of S .
4. Let Ω be the solid region inside the sphere $x^2 + y^2 + z^2 = 1$ and below the inverted cone $z = \sqrt{x^2 + y^2} - 1$.

(a) Express the volume of Ω as iterated integrals in Cartesian coordinates with the order $dzdyzx$,

(b) Express the volume of Ω as iterated integrals in Cartesian coordinates with the order $dydzdx$,

(c) Express the volume of Ω as iterated integrals in cylindrical coordinates.

(d) Let Ω' be region inside the same sphere but below the cone $z = \sqrt{x^2 + y^2}$. Express the volume of Ω' as iterated integrals with an appropriate coordinate system and an order of integration.

5. Let E be the solid region bounded by the surfaces

$$(x - y)^2 + (y - z)^2 = 1, \quad x + z = -(x - y)^2 - (y - z)^2, \quad x + z = \sqrt{(x - y)^2 + (y - z)^2}.$$

Express the volume of E as iterated integrals.

Math 213 Recitation - Week 05

1. Let S be the solid region in $\mathbb{R}^3$ bounded by the surfaces $x^2 + y^2 = 1$, $z = -x^2 - y^2$, and $z = \sqrt{x^2 + y^2}$.

- (a) Express the volume of S as iterated integrals in spherical coordinates.
- (b) Compute the volume of S using the cylindrical coordinates.

2. Evaluate

$$\int_{-1}^1 \int_0^{\sqrt{1-y^2}} \int_0^x (x^2 + y^2) dz dx dy$$

using cylindrical coordinates.

3. Consider the region E in $\mathbb{R}^3$ bounded by the surfaces $z = \sqrt{x^2 + y^2}$ and $z = x^2 + y^2$. Express the volume of E as iterated triple integrals

- (a) in Cartesian coordinates,
- (b) in cylindrical coordinates,
- (c) in spherical coordinates.
- (d) Also express the volume of E as an iterated double integral.

4. Find the volume of the solid lying inside both the sphere $x^2 + y^2 + z^2 = 2a^2$ and the cylinder $x^2 + y^2 = a^2$.

5. Consider

$$\iiint_T x^2 dV$$

where T is the region bounded by the vertical cylinders of radius 3 and 4, and the planes $z = 0$ and $z = 3$.

- (a) Express the same integral as iterated integrals in Cartesian coordinates.
 - (b) Compute the triple integral using cylindrical coordinates.
 - (c) Express the same integral as a double integral.
6. Let T be the solid region in the unit sphere and bounded below by the paraboloid $z = x^2 + y^2$.
- (a) Express the volume of T in cylindrical and spherical coordinates.
 - (b) Express the volume of T as an iterated double integral.
7. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) = x^2 + y^2$. Let $D = [-1, 1] \times [-1, 1]$. Let $\mathcal{P}$ be the partition of D into four sub-squares of equal area. Find $\mathcal{L}(f, \mathcal{P})$ and $\mathcal{U}(f, \mathcal{P})$.

8. Evaluate

$$\iint_D \frac{6}{x} y^2 \sin(x^2 y^2) dA$$

where D is the region bounded by the curves $xy = 1$, $xy = 2$, $y = x^2$ and $y = 3x^2$.

9. Evaluate

$$\int_0^{\pi/2} \int_0^{(\cos \theta + \sin \theta)^{-1}} r^2 \cos \theta dr d\theta.$$

10. Let $D = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq y \leq x\}$. Find

$$\iint_D x e^{-\frac{x^2}{y}} dA$$

if it exists.

Math 213 Recitation - Week 06

1. Parametrize the curve C where

- (a) C is the arc on the circle $(x - 2)^2 + y^2 = 4$ starting at the point $(4, 0)$ and ending at the point $(2, 2)$ oriented clockwise.
- (b) C is the ellipse which is the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = 2y + 1$, and oriented counter-clockwise as viewed from above.
- (c) C is the curve of intersection of the surfaces $z = \sqrt{1 - x^2 - y^2}$ and $x + y = 1$. (orientation doesn't matter)
- (d) C is the curve of intersection of the paraboloid $y = 4 - x^2 - z^2$ and the parabolic cylinder $y = 4 - 2x^2$. (orientation doesn't matter)

2. Find the length of the curve given by

$$r(t) = (\cos t + t \sin t)\mathbf{i} + (\sin t - t \cos t)\mathbf{j}, \quad -\pi/2 \leq t \leq \pi.$$

3. Evaluate the following line integrals:

- (a) $\int_C (x - y + 2z)ds$ where $C : r(t) = (1 + t, 2t, 1 - t)$, $t \in [0, 1]$.
 - (b) $\int_C (x^2 + y^2 + z)ds$ where $C : r(t) = (\cos t, \sin t, t)$, $t \in [0, 1]$.
 - (c) $\int_C 3ze^{x^2}ds$ where C is the line segment from $(0, 0, 0)$ to $(2, 1, 3)$.
 - (d) $\int_C \sqrt{z}ds$ where $C : r(t) = (2 \cos t, 2 \sin t, t^2)$, $t \in [0, 1]$.
4. Express the arc-length of the part of the Archimedean spiral $r = \theta$ where $\theta \in [0, 4\pi]$ as a line integral.
5. Compute the surface area of the surface given by

$$S = \{(x, y, z) \mid x^2 + y^2 = 4, \ 0 \leq z \leq x + y^2\}$$

as a line integral.

Math 213 Recitation - Week 07

1. Evaluate

$$\int_C (x dx + x^2 y dy)$$

where C is the part of the circle $x^2 + y^2 = 4$ from $(-2, 0)$ to $(0, 2)$ followed by the line segment from $(0, 2)$ to $(0, 3)$.

2. Evaluate $\int_C (x^2 y dx + x^3 y^2 dy)$ where C is the closed curve formed by the portions of the line $y = 4$ and the parabola $y = x^2$ oriented counter-clockwise.
3. Let $F(x, y, z) = (yz + e^x, 2xz + 2y, xy + \sin z)$. Evaluate

$$\int_C F \cdot dr$$

where C is the curve parametrized as $r(t) = (t, t^2, \pi t^3)$, $t \in [0, 1]$.

4. Let C be the ellipse formed by the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = 2y + 1$, oriented counter-clockwise as viewed from above, and let $F(x, y, z) = (y, z, x)$. Evaluate $\int_C F \cdot dr$.
5. Find a potential ϕ to show that the vector field

$$F(x, y) = (\arctan y - ye^{xy}, \frac{x}{1+y^2} - xe^{xy} + 6y)$$

is conservative.

6. Find a potential ϕ to show that the vector field

$$F(x, y, z) = (\frac{-2x}{x^2 + y^2}, z \cos(yz) - \frac{2y}{x^2 + y^2}, y \cos(yz) + 12z^2)$$

is conservative.

7. Show that the vector field

$$F(x, y) = (y \cos(xy) + x, x \cos(xy) + x)$$

is **not** conservative.

8. Show that the vector field

$$F(x, y, z) = (yze^{xyz} + \cos x + y \arctan z, xze^{xyz} + \arctan z + z, xye^{xyz} + \frac{xy}{1+z^2} + 2y)$$

is not conservative.

9. Let $F(x, y, z) = (2ax \sin y + 2byz, ax^2 \cos y + b^2xz, (a-b-1)x + (b+1)e^z)$ where a, b are real parameters. Find the values of a and b such that F is conservative, and find a potential.

Math 213 Recitation - Week 08

1. Consider the vector field

$$F(x, y, z) = (2xz^3 + 2xy \cos(x^2y), \frac{z}{1+y^2} + x^2 \cos(x^2y), \arctan y + 3x^2z^2 + 1).$$

- (a) Show that F is conservative.
- (b) Argue that the line integrals of F are path-independent.
- (c) Compute $\int_C F \cdot dr$ where

$$C : r(t) = (\sqrt{\pi}t^{1/3} + \sin(t - t^2), t^2, \sin(\frac{\pi}{2}t)), \quad t \in [0, 1]$$

using a more suitable curve.

- (d) Compute the integral in part (c) by finding a potential function for F .
- (e) Let C_2 be the curve of intersection of the ellipsoid $2x^2 + 5y^2 + 3z^2 = 1$ and the plane $y + 2x = 0$. Compute

$$\int_{C_2} F \cdot dr$$

in your head.

2. Evaluate

$$\int_C (x \sin(y^2) - y^2)dx + (x^2y \cos(y^2) + 3x)dy$$

where C is the CCW boundary of the trapezoid with vertices $(0, -2)$, $(1, -1)$, $(1, 1)$ and $(0, 2)$.

3. If C is the intersection of the surfaces $z = \ln(1 + x)$ and $y = x$ from $(0, 0, 0)$ to $(1, 1, \ln 2)$, evaluate

$$\int_C (2x \sin(\pi y) - e^z)dx + (\pi x^2 \cos(\pi y) - 3e^z)dy - xe^z dz.$$

4. Consider the vector field $F(x, y) = \frac{-y\mathbf{i} + x\mathbf{j}}{x^2 + y^2}$.

- (a) Find a potential function ϕ for F in a domain D .
- (b) By sketching F , make a guess as to whether the line integrals of F are always path independent.
- (c) Prove that line integrals of F are not always path independent.
- (d) Explain why we do not have path-independent-ness even though there is a potential function.
- (e) Find a domain D such that the line integrals of F are always path-independent for every curve lying in D .
- (f) Find $\int_C F \cdot dr$ where C is any piecewise smooth curve going from $(1, 1)$ to $(2, 2\sqrt{3})$ lying in $\{(x, y) \in \mathbb{R}^2 \mid x > 0\}$ using the potential ϕ .
- (g) Show that for any curve C missing the origin and lying in the line $y = x$, $\int_C F \cdot dr = 0$ geometrically.
- (h) Explain why we cannot use the same method as in part (f) for computing $\int_C F \cdot dr$ where C is a curve going from $(-1, 1)$ to $(1, 1)$.

Math 213 Recitation - Week 09

1. Evaluate

$$\oint_C (x - y^3)dx + (y^3 + x^3)dy$$

where C is the upper-right quarter disk of radius 2 oriented CCW.

2. Let C be the curve consisting of the parts of the curves $y = 1/x$, $y = 4/x$, $y = x$ having the initial and the end point as $(1/2, 2)$ and $(1, 4)$, respectively. Use Green's Theorem suitably to evaluate

$$\int_C \frac{y^2}{x} dx + y \ln x dy.$$

3. Find

$$\oint_C \frac{y dx - (x - 1) dy}{(x - 1)^2 + y^2}$$

if C is the curve $x^2 + y^2 = 4$ oriented CCW.

4. Let C be the positively oriented, simple, closed curve in the xy -plane bounding a region D and not passing through the origin. Show that

$$\oint_C \frac{-y dx + x dy}{x^2 + y^2} = \begin{cases} 0, & \text{if the origin is outside } D \\ 2\pi, & \text{if the origin is inside } D. \end{cases}$$

5. Use Green's Theorem to find the area of the elliptic disk $x^2 + 4y^2 \leq 4$.

6. Evaluate

$$\oint_C \frac{x\mathbf{i} + y\mathbf{j}}{x^2 + y^2} \cdot d\mathbf{r}$$

where C is the positively oriented boundary of the square with vertices $(1, 0)$, $(0, 1)$, $(-1, 0)$ and $(0, -1)$.

7. Consider the surface given by the parametrization

$$\sigma(u, v) = (u, v^2, u^2 + v).$$

Find the equation of the tangent plane at the point $\sigma(1, 1)$.

8. Parametrize the following surfaces.

- (a) The elliptical paraboloid $z = 4 - x^2 - y^2$ for $z \geq 0$, in both orientations.
- (b) The upper hemisphere of radius 3 centered at the origin, but only the part in the first octant, in both orientations.
- (c) The portion of the cone $z = \sqrt{x^2 + y^2}$ bounded between the planes $z = 1$ and $z = 4$, oriented upwards.
- (d) The part of the plane $2x - y + z = 6$ inside the cylinder $x^2 + y^2 = 9$, oriented downwards.
- (e) The part of the hyperbolic paraboloid $z = x^2 - y^2$ above the region $-1 \leq x, y \leq 1$, with any orientation if it is orientable.

Math 213 Recitation - Week 10

1. Parametrize the following surfaces.

- (a) The elliptical paraboloid $z = 4 - x^2 - y^2$ for $z \geq 0$
- (b) The upper hemisphere of radius 3 centered at the origin
- (c) The portion of the cone $z = \sqrt{x^2 + y^2}$ bounded between the planes $z = 1$ and $z = 4$
- (d) The part of the plane $2x - y + z = 6$ inside the cylinder $x^2 + y^2 = 9$
- (e) The part of the hyperbolic paraboloid $z = x^2 - y^2$ above the region $-1 \leq x, y \leq 1$

2. Consider the surface given by the parametrization

$$\sigma(u, v) = (u, v^2, u^2 + v).$$

Find the equation of the tangent plane at the point $\sigma(1, 1)$.

- 3. Find the area of the part of the surface $z = x^2 + y^2$ inside the cylinder $x^2 + y^2 = 1$ using surface integral.
- 4. Express the area of the ellipsoid $x^2/4 + y^2/9 + z^2/16 = 1$ as a surface integral.
- 5. Find the area of the part of the cylinder $x^2 + y^2 = 2y$ lying inside the sphere $x^2 + y^2 + z^2 = 4$.
- 6. Evaluate

$$\iint_S (x - y + z) dS$$

where S is the surface of the solid enclosed by $x^2 + y^2 = 1$, $z = x + y + 2$, $z = x^2 + y^2 - 1$.

7. Find a piecewise smooth, simple closed curve C that is oriented CCW such that

$$\int_C (y + e^x) dx + (2x + e^y) dy = 4.$$

Hint: Use Green's Theorem.

Review

8. Evaluate

$$\oint_C \frac{x\mathbf{i} + y\mathbf{j}}{x^2 + y^2} \cdot d\mathbf{r}$$

where C is the positively oriented boundary of the square with vertices $(1, 0)$, $(0, 1)$, $(-1, 0)$ and $(0, -1)$.

- 9. **(Quiz 3)** Let $F(x, y) = (\frac{-y}{x^2}, \frac{1}{x})$, and let C be the part of the curve $y = e^{x^2}$ from $(1, e)$ to $(2, e^4)$. Evaluate

$$\int_C F \cdot d\mathbf{r}.$$

- 10. Evaluate the integral in Problem 8 without using Green's theorem for the curve C where C is the square with vertices $(1, 1)$, $(-1, 1)$, $(-1, -1)$ and $(1, -1)$.

Math 213 Recitation - Week 11

1. Evaluate

$$\iint_S (y^2, z^2, x^2) \cdot d\mathbf{S}$$

where S is the portion of the paraboloid $z = 4 - x^2 - y^2$ between $z = 2$ and $z = 4$, oriented upwards.

2. Evaluate

$$\iint_S xdydz + ydzdx + zdx dy \quad \partial \vec{S} = (\overset{=dydz}{\partial A_x}, \overset{=zdx}{\partial A_y}, \overset{=xdy}{\partial A_z}),$$

where S is the cylinder $x^2 + y^2 = 1$ for $0 \leq z \leq 3$, oriented outwards. where ∂A_x is the

3. Evaluate

$$\iint_S xdydz - ydzdx + zdx dy$$

one element projected on yz -plane and so on.

where S is the plane passing through the points $(0, 0, 1)$, $(1, 1, 0)$ and $(0, 2, 2)$, oriented upwards.

4. Calculate $\nabla \cdot F$ and $\nabla \times F$ for the following vector fields:

(a) $F = xz^2\mathbf{i} + y^2z\mathbf{j} + xyz\mathbf{k}$,

(b) $F(r, \theta) = r^2 \sin^2 \theta \mathbf{i} + r \cos \theta \mathbf{j}$.

5. If the vector fields F and G are smooth and conservative, show that $F \times G$ is solenoidal. Also, find a vector potential for $F \times G$.
6. Let F be an irrotational, solenoidal, smooth vector field on $\mathbb{R}^3$. Using the identity

$$\nabla^2 F = \nabla(\nabla \cdot F) - \nabla \times (\nabla \times F),$$

show that the three components of F and any scalar potential for F are all harmonic functions in $\mathbb{R}^3$.

7. Let $\vec{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, and let $u = |\vec{r}|$. If f is a differentiable function of one variable, show that

$$\nabla \times (f(u)\vec{r}) = \vec{0}.$$

Math 213 Recitation - Week 12

1. Find a vector potential for $\mathbf{F}(x, y, z) = (-y, x, 0)$.
2. Compute

$$\iint_S (x, y, z) \cdot d\mathbf{S}$$

where S is the surface obtained by rotating the curve

$$C : x = \cos z, \ y = 0; \ z \in [0, \pi]$$

about the z -axis by 2π , oriented outwards.

3. Let $\mathbf{F}(x, y, z) = (x^2 + y^2 + z^2)(x\mathbf{i} + y\mathbf{j} + z\mathbf{k})$, and let S be the sphere of radius a about the origin, oriented outwards. Compute

$$\iint_S \mathbf{F} \cdot d\mathbf{S}$$

both directly and by the divergence theorem.

4. Let R be a regular region in $\mathbb{R}^3$ with piecewise smooth boundary. Find a vector field $\mathbf{F}$ such that the surface integral

$$\iint_{\partial R} \mathbf{F} \cdot d\mathbf{S}$$

computes the volume of R , where ∂R is oriented outwards.

5. Compute

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{N} dS$$

where S is the surface $x^2 + y^2 + (z - 1)^2 = 6$, $z \geq 0$, N is the unit outward normal on S , and

$$\mathbf{F}(x, y, z) = (xy - y^3 \cos z, x^3 e^z, xyz e^{x^2 + y^2 + z^2}).$$

6. Compute

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

where C is the curve

$$\mathbf{r}(t) = (\cos t, \sin t, \sin 2t), \ t \in [0, 2\pi]$$

and $\mathbf{F}(x, y, z) = (e^x - y^3, e^y + x^3, e^z)$.

Hint: Show that C lies on the surface $z = 2xy$.

7. Compute

$$\int_C y^2 dx + z^2 dy + x^2 dz$$

where C is the boundary of the triangle with vertices $(1, 0, 0)$, $(0, 2, 0)$ and $(0, 0, 3)$, oriented CCW when looked down from a height.

Math 213 Recitation - Week 13

1. Given the transformation

$$x = u \cos v, \quad y = u \sin v, \quad z = w;$$

- (a) Identify the coordinate system,
- (b) Describe the coordinate surfaces $u = \text{constant}$, $v = \text{constant}$ and $w = \text{constant}$.

2. Given the transformation

$$x = u \cos v, \quad y = u \sin v \sin w, \quad z = u \sin v \cos w;$$

for nonsingular points,

- (a) Compute the scale factors h_u , h_v and h_w ,
 - (b) Find the local basis vectors $\hat{u}$, $\hat{v}$ and $\hat{w}$,
 - (c) Determine whether the coordinate system is orthogonal.
3. Let $f(u, v, w) = u^2 + v^2 + w^2$ in the coordinate system of Problem 2.
- (a) Express ∇f in terms of local basis vectors.
 - (b) Evaluate ∇f at the point $(u, v, w) = (1, 2, \pi/2)$.

4. Do the following.

- (a) Calculate ∇f where $f(r, \theta, z) = r\theta z$.
- (b) Calculate ∇f where $f(\rho, \phi, \theta) = \rho\phi\theta$.
- (c) Calculate $\nabla \cdot \mathbf{F}$ where $\mathbf{F}(\rho, \phi, \theta) = (\sin \phi, 0, 0)$.
- (d) Calculate $\nabla \times \mathbf{F}$ where $\mathbf{F}(r, \theta, z) = (0, r, 0)$.

5. Express $\nabla f(r, \theta)$, $\nabla \cdot \mathbf{F}(r, \theta)$, $\nabla \times \mathbf{F}(r, \theta)$ in terms of (plane) polar coordinates (r, θ) .

6. Using the identity $\nabla^2 f = \nabla \cdot (\nabla f)$; calculate

- (a) $\nabla^2 f(r, \theta, z)$ in terms of cylindrical coordinates,
- (b) $\nabla^2 f(\rho, \phi, \theta)$ in terms of spherical coordinates,
- (c) $\nabla^2 f(u, v, w)$ in terms of arbitrary orthogonal curvilinear coordinates (u, v, w) .

Review of Div. and Stokes

7. Let S be the part of the paraboloid $z = x^2 + y^2$ from $z = 4$ to $z = 9$, oriented upwards. Compute

$$\iint_S (x^3, y^3, z^3) \cdot d\mathbf{S}$$

using Divergence Theorem.

8. Let C be the curve given by the parametrization $\mathbf{r}(t) = (t, 0, t^2)$, $t \in [-1, 1]$. Compute

$$\int_C (-yz, xy, xz) \cdot d\mathbf{r}$$

using Stokes' Theorem.

Math 213 Recitation - Week 14

1. Evaluate

$$\oint_C (e^x \sin y + 3y)dx + (e^x \cos y + 2x - 2y)dy$$

where C is the ellipse $4x^2 + y^2 = 4$, oriented CCW.

2. Find a positively oriented, piecewise smooth, simple, close curve C such that

$$\oint_C (y + xy^2)dx + (3x + x^2y)dy = 8\pi.$$

3. Find the area of the torus obtained by rotating the circle $(x - 2)^2 + z^2 = 1$ about the z -axis.

4. Evaluate

$$\iint_S \mathbf{F} \cdot d\mathbf{S}$$

where $\mathbf{F}(x, y, z) = (x, y, z)$, and S is the boundary of the region $x^2 + y^2 \leq z \leq \sqrt{2 - x^2 - y^2}$, oriented outward.

5. Find a vector potential for

$$\mathbf{F}(x, y, z) = (xy + z)\mathbf{i} + xz\mathbf{j} - (yz + x)\mathbf{k}.$$

6. Calculate

$$\oiint_S \mathbf{F} \cdot d\mathbf{S}$$

where $\mathbf{F}(x, y, z) = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$ and S is surface of the cube $0 \leq x, y, z \leq a$, oriented outward.

7. Evaluate

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$$

where $\mathbf{F}(x, y, z) = (y, x - 2x^3z, xy^3)$ and S is the upper half of the sphere $x^2 + y^2 + z^2 = a^2$, oriented outward.

8. Let S be the portion of the paraboloid $z = x^2 + y^2$ below the plane $2x + 2y + z = 2$, oriented downward. Compute

$$\iint_S (x, -y, xy) \cdot d\mathbf{S}$$

using Stokes' Theorem.

9. Given any non-vertical plane P parallel to x -axis, let C be the curve of intersection of P with the cylinder $x^2 + y^2 = a^2$. Show that

$$\oint_C (yz - y)dx + (xz + x)dy = 2\pi a^2.$$